Humanized PD-L1 Mouse

Strain Name: C57BL/6-*Cd274* $^{em1(hPD\text{-}L1)/Smoc}$ Strain Background: C57BL/6 Cat. No. : IT-HU-00062

Programmed cell death 1 ligand (PD-L1), also known as cluster of differentiation 274(CD274) or B7 homolog 1 (B7H1), is a 40 kDa transmembrane protein encoded by the gene CD274 in human. The binding of PD-L1 to the PD-1 receptors expressed on the surface of activated T cells transmits a negative regulatory signal. While under normal circumstances the PD-L1 pathway acts as a type of "off switch" that helps keep the T cells from attacking other cells, the high expression level of PD-L1 in the tumor microenvironment inhibits the function of tumor-infiltrating T cells, thereby allowing tumors cells to escape immune surveillance.

Construction strategy

On the C57BL/6 background, the full-length coding sequence of human CD274 gene was placed immediately downstream of the start codon of the mouse endogenous Cd274 gene, followed by a poly(A) site. This guarantees an exclusive expression of human PD-L1 in the humanized PD-L1 mice.

Validation data

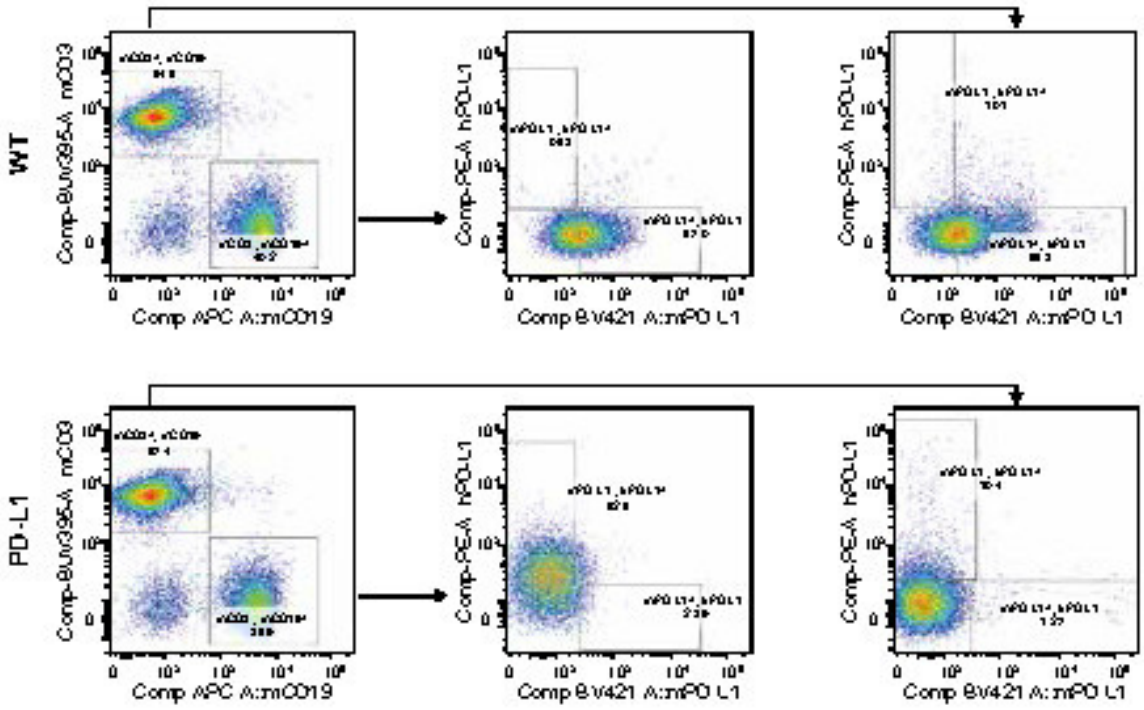
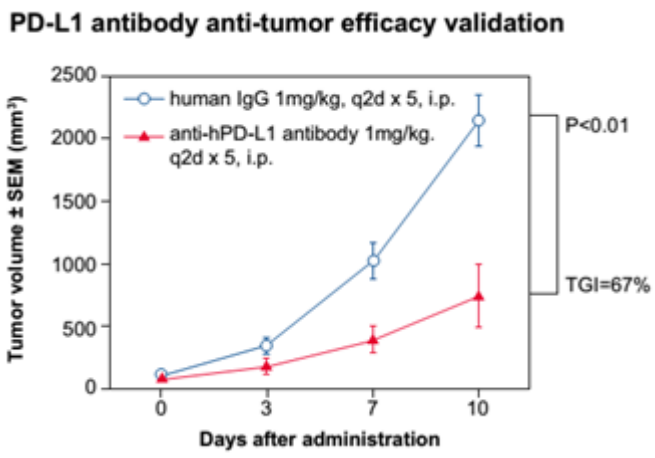


Figure 1. Expression of PD-L1 in the spleen lymphocytes collected from homozygous humanized PD-L1 mice and wild-type mice is detected by FACS. The results showed that the expression of human PD-L1 can be detected in both T cells and B cells collected from the spleen of homozygous humanized PD-L1 mice. (Completed in collaboration with CrownBio)



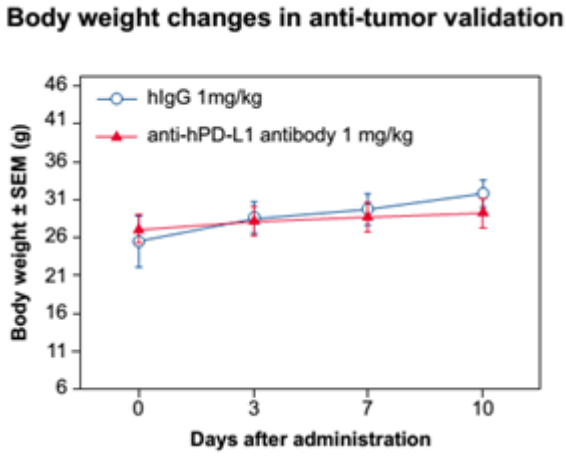


Figure 2. In vivo validation of anti-tumor efficacy in a MC38 tumor-bearing model of humanized PD-L1 mice. Homozygous humanized PD-L1 mice were inoculated with MC38 colon cancer cells (expressing human PDL1 rather than murine PD-L1). After the tumors grew to 100 mm^3, the animals were randomly assigned into a control group and a treatment group (n=5). The results showed: The antibodies targeting human PD-L1 were associated with a very significant anti-tumor effect (TGI: tumor growth inhibition, $p < 0.001$), demonstrating that the humanized PD-L1 mice are a good in vivo model for validating the efficacy of antibodies targeting human PD-L1.

Humanized OX40 Mouse

Strain Name: C57BL/6-*Tnfrsf4* $^{em1(hTNFRSF4)Smoc}$ Strain Background: C57BL/6 Cat. No. : IT-HU-00041

OX40 is a co-stimulatory molecule expressed on the surface of activated cytotoxic T cells and regulatory T cells. Administration of agonistic, anti-OX40 antibody increases proliferation of peripheral blood $CD4^+$ and $CD8^+$ T cells, thereby creating a tumor microenvironment that is more favorable to anti-tumor immune responses. Accumulating preclinical evidence supports the application value of anti-OX40 antibodies in cancer therapy, and several such agonistic antibodies are now tested in early stage of clinical trials. The humanized OX40 mice developed by SMOC provide a translational model that enables the *in vivo* efficacy evaluation of human-specific therapeutic OX40 antibodies.

Construction strategy

The humanized OX40 mouse model was developed on the C57BL/6 background. A chimeric expression cassette that encodes the extracellular domain of human OX40 as well as the transmembrane and intracellular domains of mouse OX40 was placed immediately downstream of the start codon of the mouse endogenous OX40 gene, followed by a poly(A) signal. Thereby, the extracellular domain of the mouse OX40 was replaced by its human counterpart while the rest of the mouse gene remained untouched.

Validation data

- Flow cytometry (FACS) analysis data of lymph node T-cells collected from humanized OX40 mice

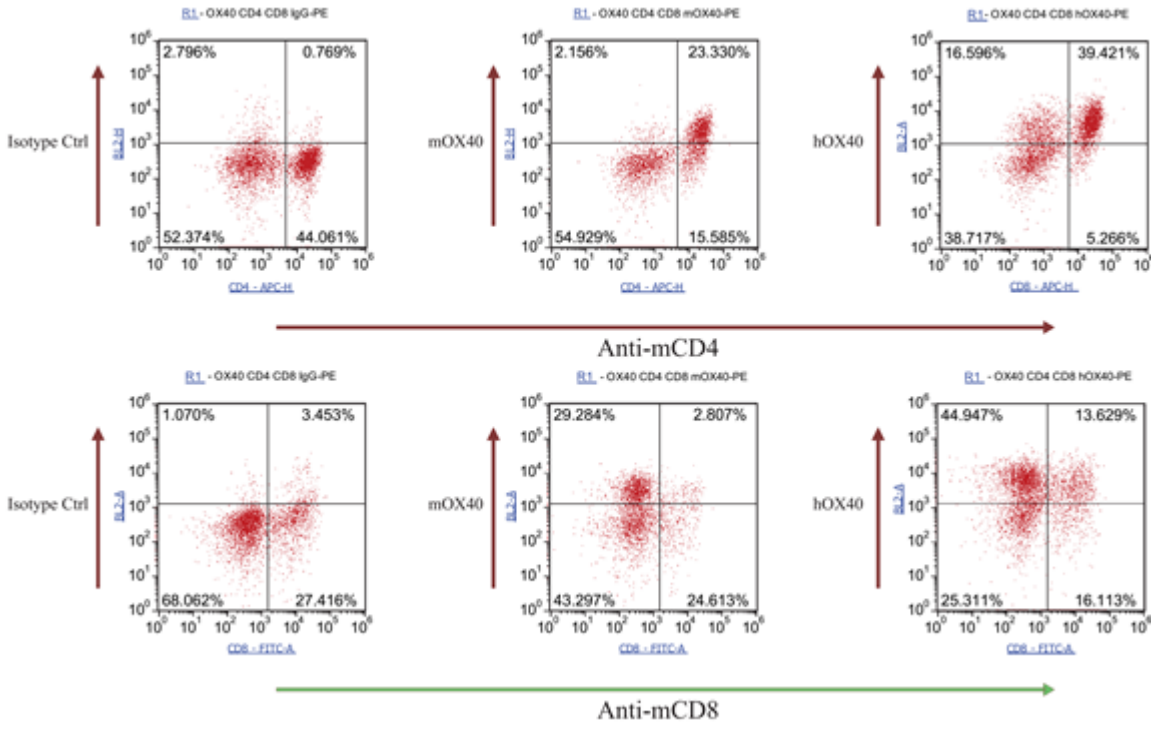
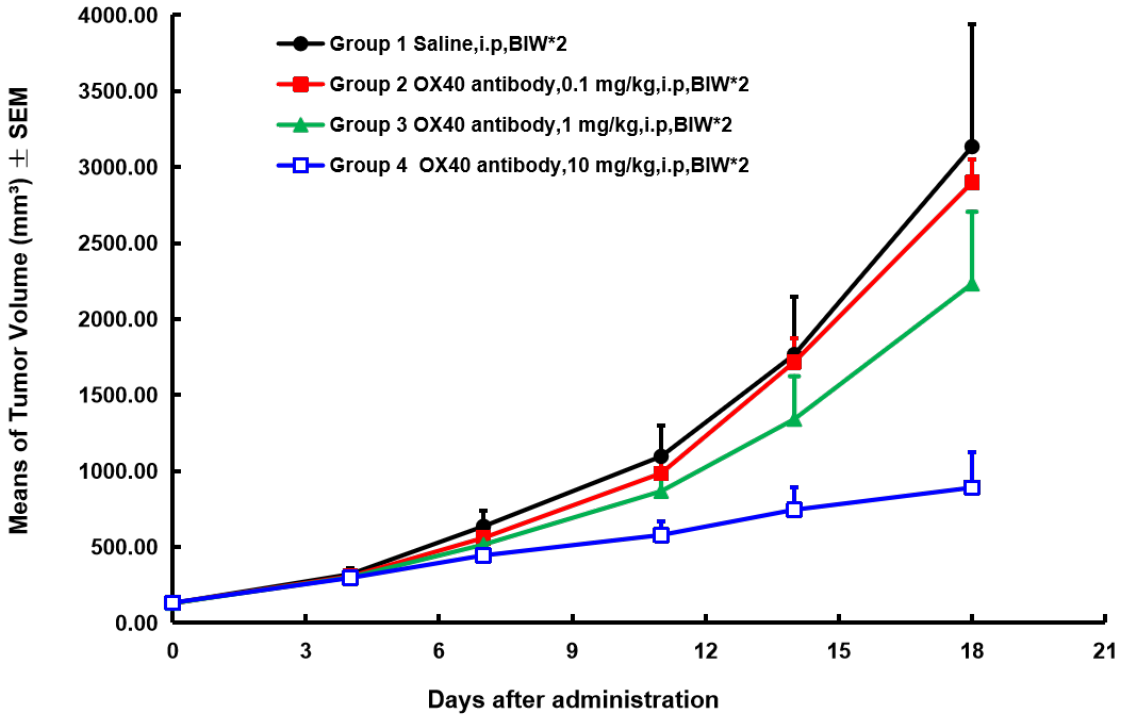


Figure 1. Expression of OX40 in the spleen lymphocytes of humanized OX40 mice is detected by FACS.

The spleen lymphocytes of heterozygous humanized OX40 mice were activated by anti-CD3 and anti-CD28 for 48 hours, and then collected for staining. Along with a group undergoing no stimulation, the expression of murine and human OX40 was detected by FACS. The results showed that the active expression of human OX40 can be detected in both activated CD4+ and CD8+ T lymphocytes collected from heterozygous humanized OX40 mice, and the expression trend of human OX40 and murine Ox40 was similar.

- In vivo validation in a MC38 tumor-bearing model of humanized OX40 mouse.

  - Case study 1



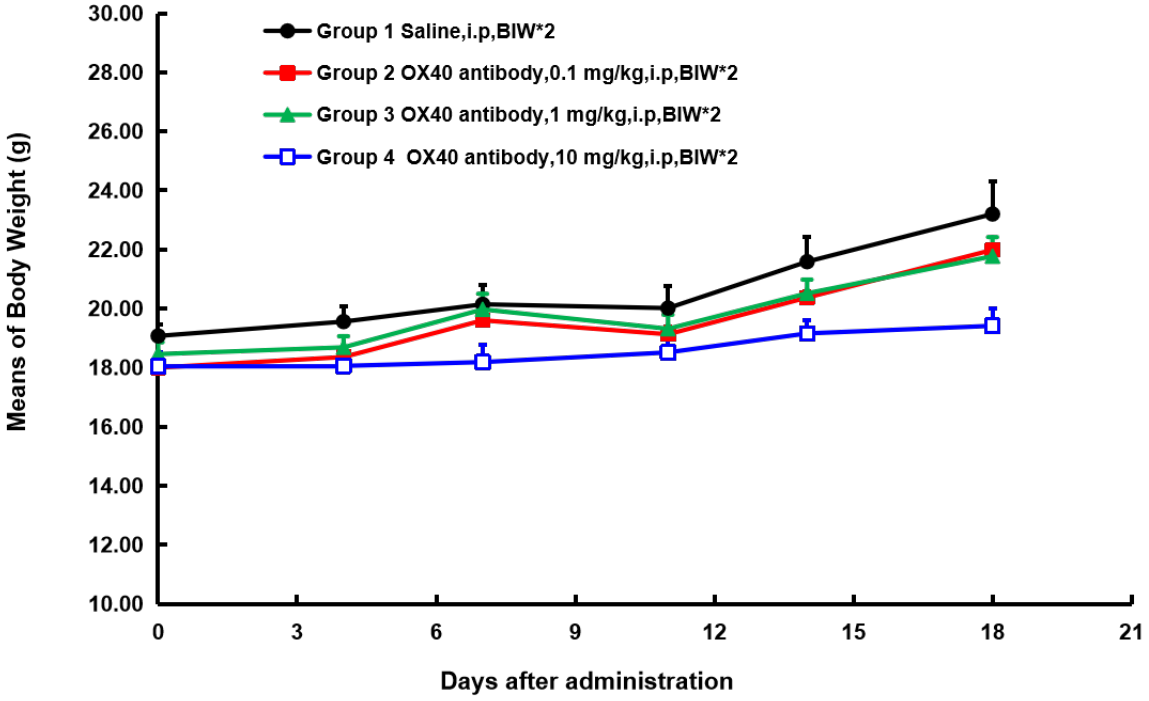


Figure 2. OX40 antibody showed dose-dependent anti-tumor activity in human OX40 knock-in mice bearing MC38 tumors. The OX40 antibody was obtained from Innoventbio.

- Case study 2

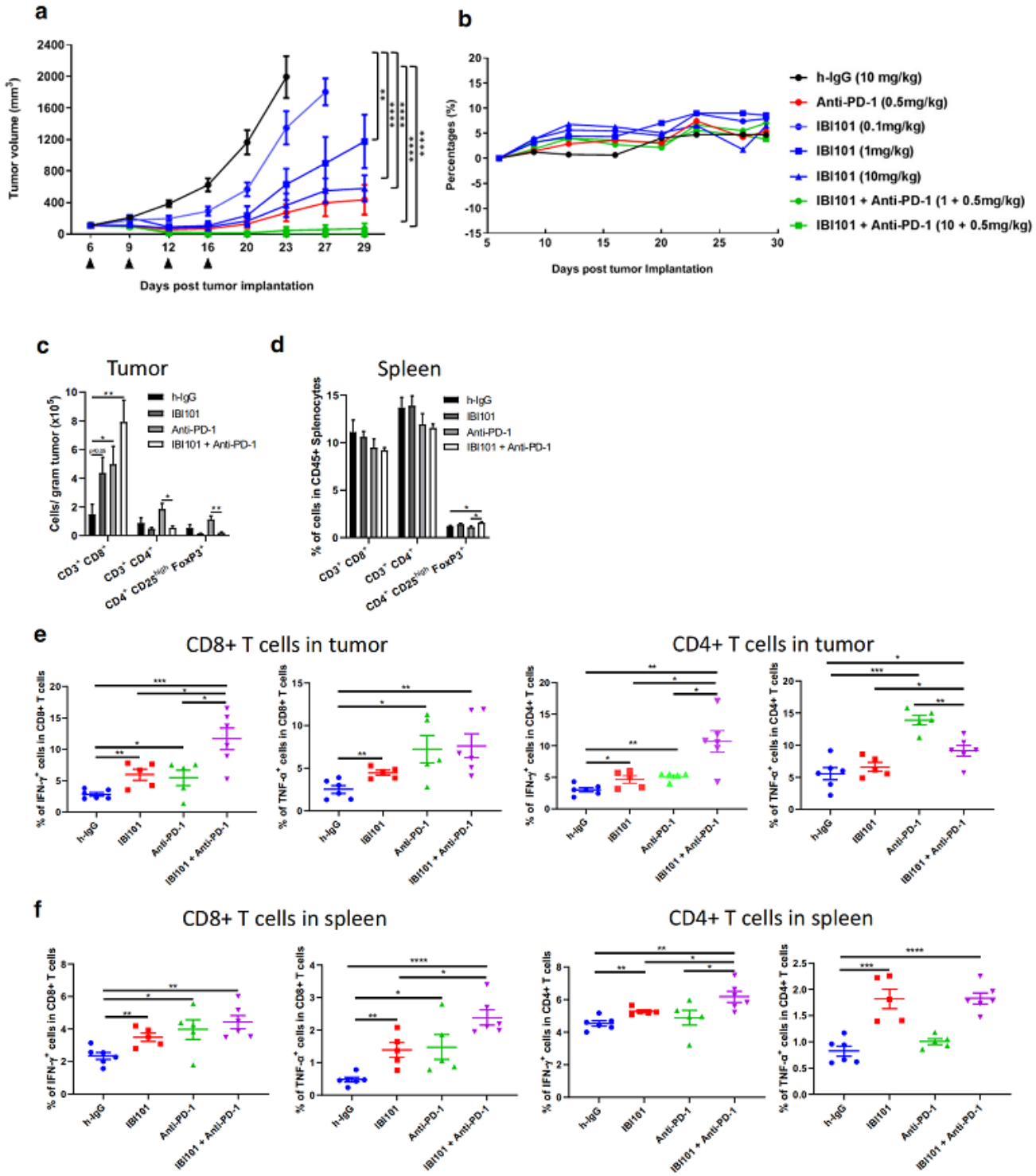

a
Tumor volume (mm³)
Days post tumor implantation
b
Percentages (%)
Days post tumor Implantation
h-IgG (10 mg/kg)
Anti-PD-1 (0.5mg/kg)
IBI101 (0.1mg/kg)
IBI101 (1mg/kg)
IBI101 (10mg/kg)
IBI101 + Anti-PD-1 (1 + 0.5mg/kg)
IBI101 + Anti-PD-1 (10 + 0.5mg/kg)
c
Tumor
Cells/ gram tumor (x10⁵)
h-IgG
IBI101
Anti-PD-1
IBI101 + Anti-PD-1
d
Spleen
% of cells in CD45+ Splenocytes
e
CD8+ T cells in tumor
CD4+ T cells in tumor
f
CD8+ T cells in spleen
CD4+ T cells in spleen
h-IgG
IBI101
Anti-PD-1
IBI101 + Anti-PD-1

Figure 3. IBI101 showed dose-dependent anti-tumor activity and enhanced tumor-specific CD8+ T cell response in human OX40 knock-in mice bearing MC38 tumors. a Tumor growth curve of mice treated with different doses of IBI101 alone or in combination with anti-mouse PD-1 antibody. Different doses of IBI101 and anti-mouse PD-1 antibody were administrated as indicated by the arrow heads after MC38 cells implantation. b Animal body weights were measured during the time course of the experiment. c Mice were injected with h-IgG (10 mg/kg), IBI101 (10 mg/kg), anti-PD-1 (0.5 mg/kg) alone or IBI101 (10 mg/kg)+anti-PD-1 (0.5 mg/kg) at day 10 and 14 post tumor cell implantation. At day 17, tumor and spleen were collected and analyzed by flow cytometry for the absolute counts of the indicated cell subsets in tumor and d proportions of indicated cell subsets in CD45+ splenocytes. Flow cytometry results showing the proportions of cytokine-secreting cancer-specific CD8+ and CD4+ T cells from tumor (e) and spleen (f) (n≥5) (In collaboration with Innoventbio)

Immune Checkpoint Humanized Mouse Models

Being recognized as a top scientific breakthrough in 2013, cancer immunotherapy is predicted to be one of the most promising research areas for improving patient outcomes. Although many immunotherapy breakthroughs may still lie ahead, important clinical advances have been made in the past few years for some of the deadliest cancers, reaffirming the potential of immunotherapy for many types of patients.

However, it is worth noting that drug candidates developed to interfere with human proteins may not comparably interact with their murine counterparts. It is therefore critical to develop humanized mouse models to enable in vivo efficacy evaluation of cancer immunotherapies.

Immune Checkpoint Humanized Mouse Models available at ingenious targeting laboratory

4-1BB
CD40
CD47
CD73 (NT5E)
CTLA4 (C57BL/6)
CTLA4 (BALB/c)
KDR
LAG3
OX40
OX40/CTLA4
PD-1 (C57BL/6)
PD-1 (BALB/c)
PD-1/4-1BB
PD-1/CD40
PD-1/CTLA4
PD-1/LAG3
PD-1/OX40
PD-1/PD-L1
PD-1/TIGIT
PD-1/TIM3
PD-L1
PD-L1/CTLA4
PD-L1/LAG3
PD-L1/OX40
PD-L1/TIGIT
SIRPA
SIRPA/CD47
TIGIT
TIM3 (C57BL/6)
TIM3 (BALB/c)
TNFRSF1B
And more to come!

To get to know more about these models, visit our website **www.genetargeting.com**
or contact our scientific experts at **inquiry@genetargeting.com**

About ingenious targeting laboratory

ingenious targeting laboratory (**ingenious**) has been a leading global provider of custom genetically modified mouse, rat, and rabbit models for over 20 years. As one of the very first mouse gene targeting companies, our trusted service is built on two decades' worth of successful animal model creation for investigators, organizations, and companies worldwide. Our models have been published in hundreds of journals including *Science*, *Nature*, and *Cell*, making us one of the most validated and respected production companies in the industry. We are excited to add catalog mouse models to our service repertoire by means of our collaboration with Shanghai Model Organisms Center (SMOC).

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